

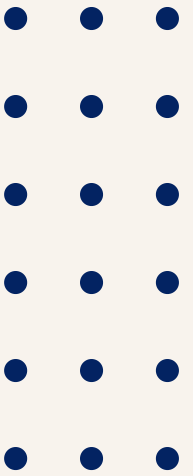
REAL TIME SENTIMENT ANALYSIS USING APACHE SPARK AND KAFKA

PRESENTED BY
Group ABC

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PROJECT BACKGROUND

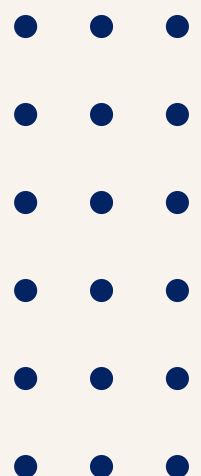


PROBLEM

- Thousands of new reviews are generated continuously.
- Manual analysis is slow, time-consuming, and impractical.
- Traditional batch processing cannot provide immediate insights.
- Businesses need real-time monitoring to respond quickly to users issues.

PROPOSED SOLUTION

Develop a real-time sentiment analysis system using Apache Kafka, Spark Structured Streaming, Elasticsearch, and Kibana to automatically classify and visualize user reviews.



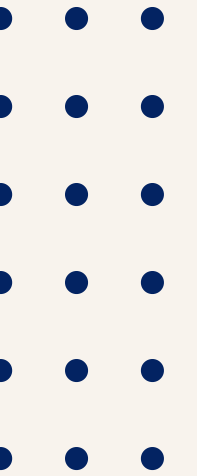


PROJECT BACKGROUND

OBJECTIVES

- Collect Google Play Store reviews for Touch 'n Go eWallet application.
- Build a sentiment classification model.
- Develop a real-time streaming pipeline.
- Visualize sentiment using Kibana.

DATASET & PREPROCESSING



RAW DATA

- Source: Google Play Store (Touch 'n Go eWallet)
- Reviews Collected: 20,000
- Collected through **google-play-scraper** library

PREPROCESSING

- Lowercase conversion
- Contraction expansion
- Malaysian slang normalization
- Remove URLs, emails, HTML tags
- Remove punctuation and emojis
- Remove stopwords
- Lemmatization

FINAL DATASET

- No of Reviews: 18835
- Attributes:
 - review_id
 - review_text
 - rating
 - thumbs_up_count
 - review_datetime
 - sentiment_label
 - cleaned_review
 - raw_review_length
 - cleaned_review_length
 - cleaned_word_count

SENTIMENT MODEL DEVELOPMENT

FEATURE ENGINEERING

- Term Frequency–Inverse Document Frequency (TF–IDF) Vectorizer for numerical feature representation

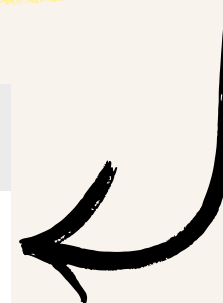
MACHINE LEARNING MODELS

- Multinomial Naive Bayes
- Linear Support Vector Machine (Linear SVM)

MODEL COMPARISON

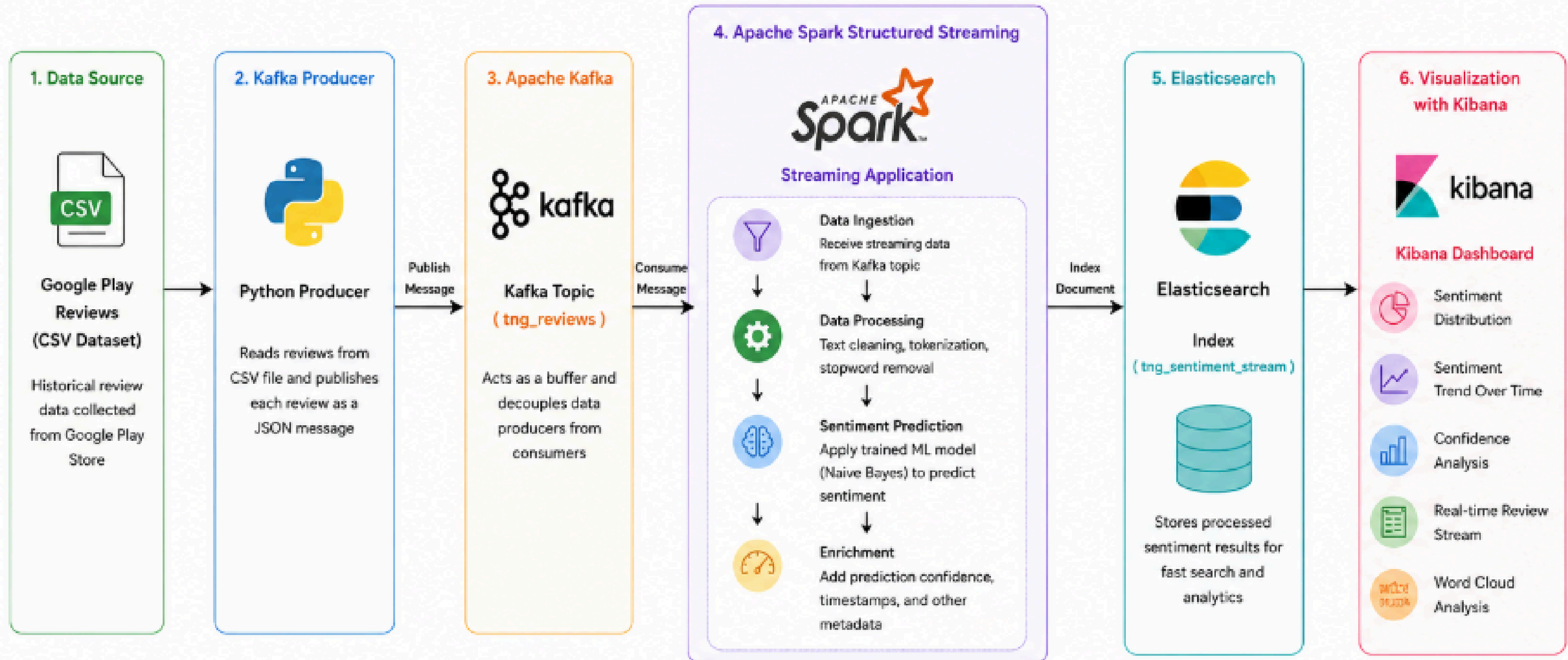
	model	accuracy	precision_weighted	recall_weighted	f1_weighted
0	Naive Bayes	0.889331	0.860148	0.889331	0.873478
1	SVM	0.868365	0.861797	0.868365	0.864344

better
performance

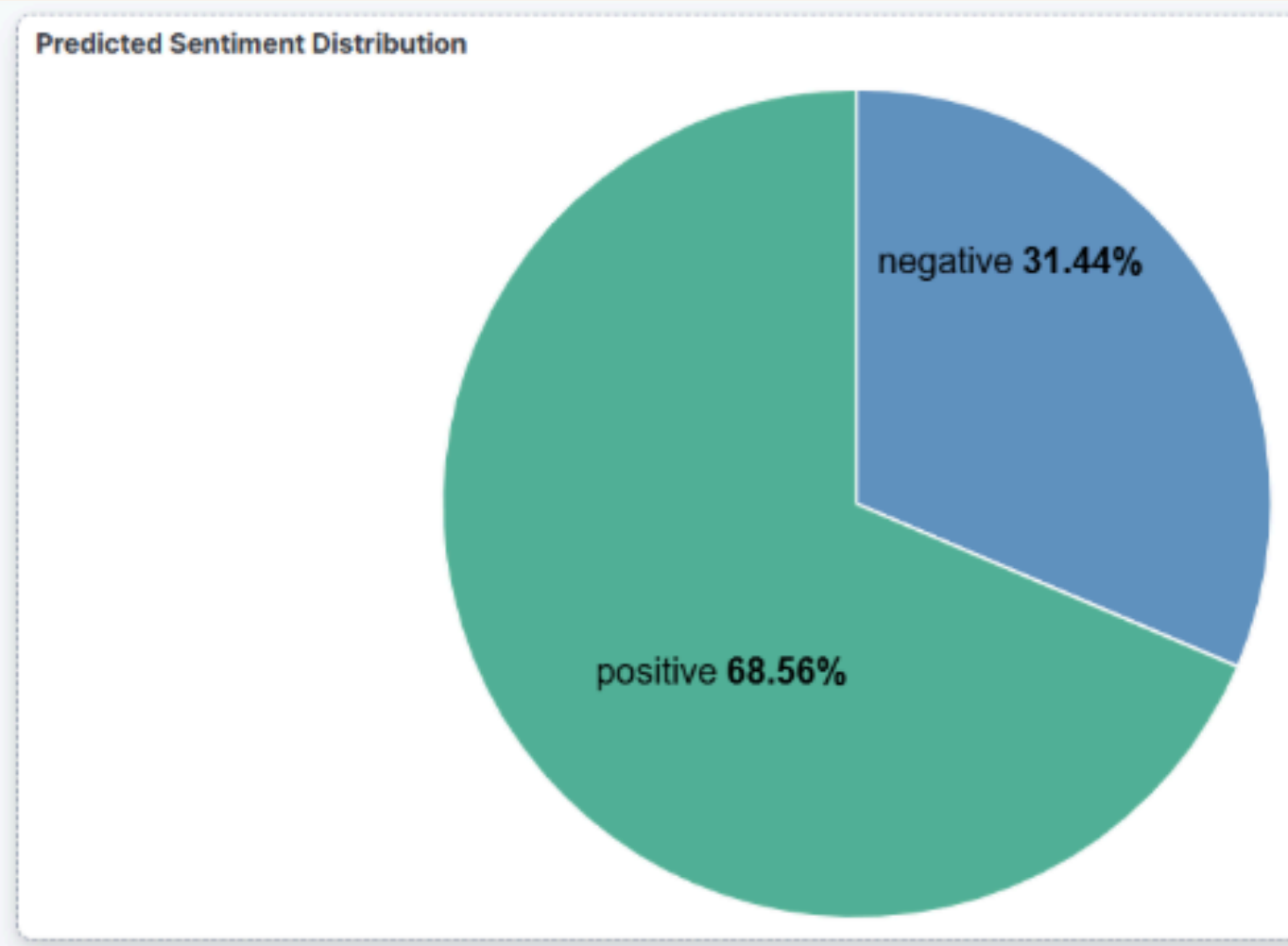
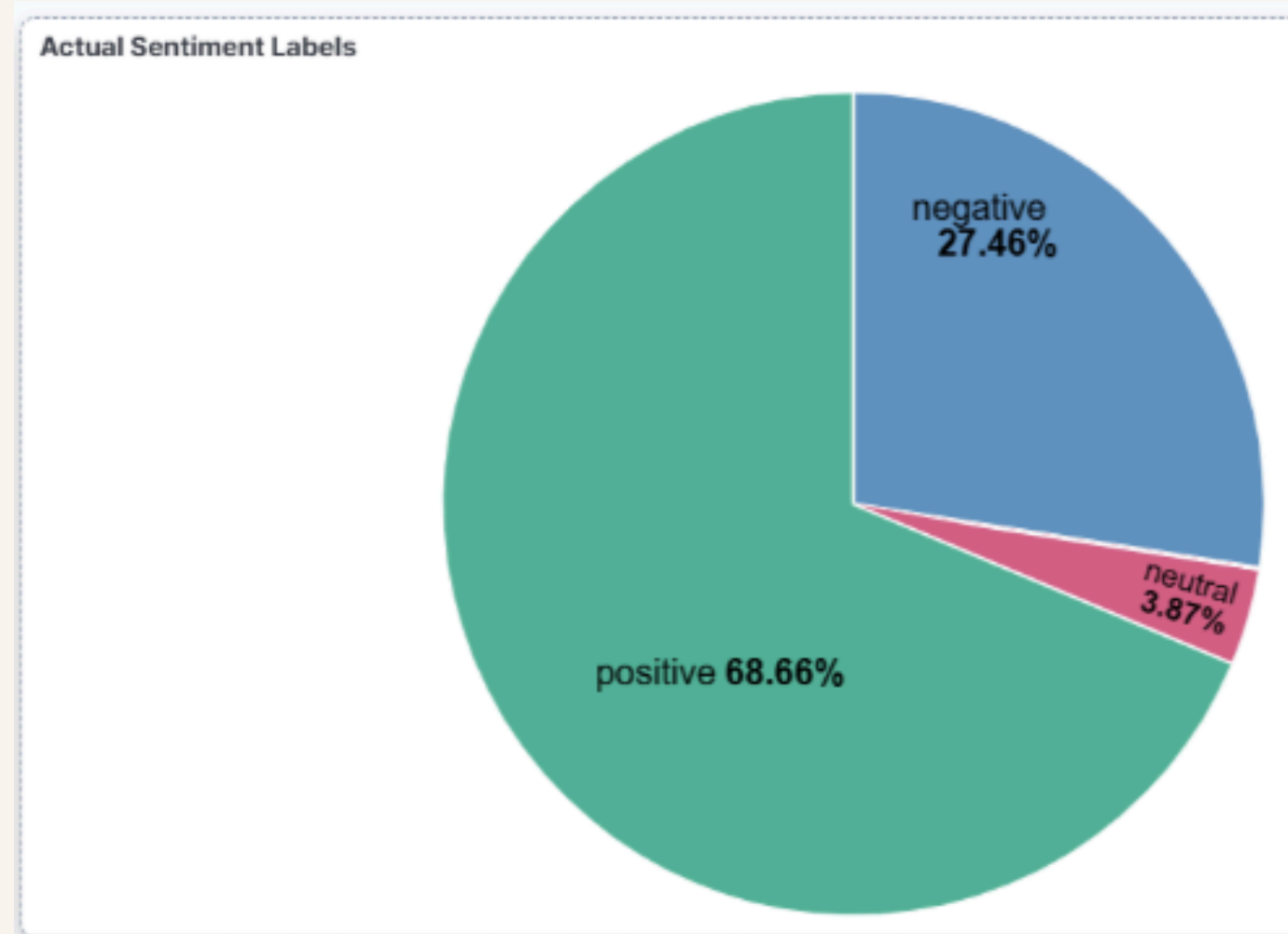


SYSTEM ARCHITECTURE

Real-Time Sentiment Analysis of TNG eWallet Reviews



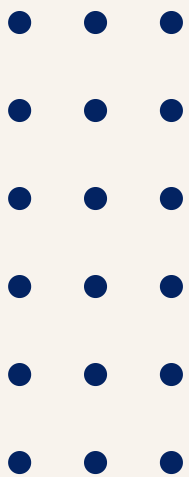
SENTIMENT ANALYSIS RESULTS



OBSERVATION

- Predicted distribution closely matches the actual data.
- Neutral reviews are mostly classified as positive or negative because they represent only a small portion of the dataset.

PREDICTION RESULTS



Prediction Outcome Summary		
Top 3 values of actual_sentiment	Top 3 values of predicted_sentim	Count of records
positive	positive	12,262
positive	negative	671
negative	negative	4,754
negative	positive	419
neutral	negative	496
neutral	positive	233

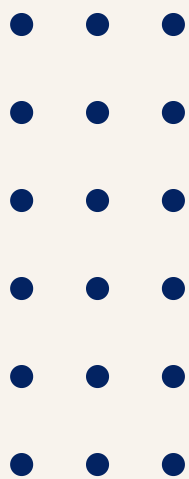
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Average Prediction Confidence

0.894

OBSERVATION

- The model demonstrates high confidence in its predictions.



REAL-TIME DASHBOARD

Real-Time Sentiment Stream

18,835 documents

Columns 5 Sort fields 1

	processed_at	review_text	predicted_sentiment	prediction_confidence	rating
☐	Jul 6, 2026 @ 17:52:18.746	Cool	positive	0.956	5
☐	Jul 6, 2026 @ 17:52:18.746	Great	positive	0.987	5
☐	Jul 6, 2026 @ 17:52:18.746	more helpful	positive	0.975	5
☐	Jul 6, 2026 @ 17:52:18.746	why cant i make a loan for the fort...	negative	0.71	1



DASHBOARD FEATURE

- Live review monitoring
- Predicted sentiment
- Prediction confidence
- Monthly sentiment trend



- Common issues:
 - Verification failures
 - Transaction problems
 - Slow response

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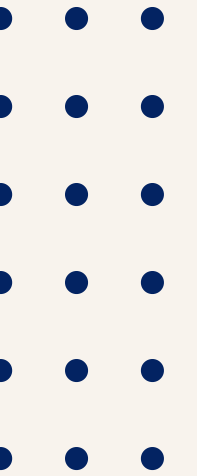
STREAMING PERFORMANCE

Performance Metric	Value
Total Records Processed	18,835
Number of Micro-Batches	97
Average Batch Size	194.18 records
Total Processing Time	949.38 seconds
Average Processing Time per Batch	9.79 seconds
Average Throughput	19.81 records/second
Overall Throughput	19.84 records/second
Minimum Throughput	15.28 records/second
Maximum Throughput	20.56 records/second

OBSERVATION

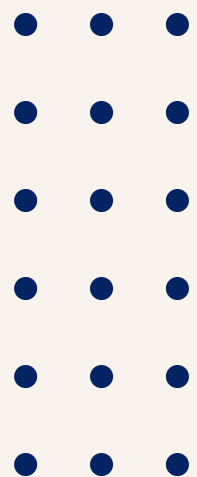
- The pipeline demonstrated consistent and reliable real-time processing performance.

CONCLUSION



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- Successfully developed a real-time sentiment analysis system.
- The selected model accurately classified customer reviews.
- Apache Kafka and Spark Streaming enabled continuous processing.
- Kibana provided interactive visualization for monitoring customer feedback.



FUTURE WORK

- Improve model performance using Grid Search or Randomized Search.
- Evaluate advanced deep learning models to improve sentiment classification accuracy.
- Expand dataset to improve model robustness and generalization.





**THANK
YOU**

